



PENETRATION TESTING REPORT

FOOTPRINTING & NETWORK SCANNING PHASES

W2-PM-FINAL | CYBERSECURITY | NETWORKWALKS

Pentester Name (Cybersecurity Professional)	Samuel Ntuen
Program/Batch	B082-Networkwalks
Date	23 August 2026
Modules completed	• W2-PM1 (Multiple Kali Reconnaissance Tools) • W2-PM2 (Google Hacking Database - GHDB) • W2-PM3 (Maltego Visual OSINT) • W2-PM4 (theHarvester Passive Discovery) • W2-PM5 (Zenmap & Nmap Subnet Scanning)
Client/Target	1. Networkwalks (secured written permission already) 2. My own local LAN Network (192.168.x.x/24)
Permission secured from client?	Yes
Phases covered	Phase 1: Reconnaissance & Footprinting Phase 2: Scanning & Network Discovery Phase 3-5: In Progress

1. Liability Disclaimer

I have performed these activities only on the systems & devices where I had secured written permission or the devices/systems that I own myself. All these materials are for education and research purpose only. Do not use anything from here to break the law. The instructor, the authors and Networkwalks are not responsible for what you do with this knowledge. Every action you take is your own responsibility. Misuse can lead to criminal charges, heavy fines, loss of your job and a permanent record. In most countries unauthorised access is a crime even when nothing is damaged.

2. Introduction

This report covers footprinting the networkwalks.com domain using six Kali Linux tools (W2-PM1), performing Google Dorking with GHDB (W2-PM2), visual OSINT mapping with Maltego (W2-PM3), passive harvesting with theHarvester (W2-PM4), and scanning my own local network with Zenmap (W2-PM5).

Together, these modules show how an attacker moves from gathering public information to mapping live hosts on a network. This is the Week 2 part of my ongoing Cybersecurity internship program at NetworkWalks Academy under the mentorship of Waqas Karim CCIE.

All commands were running in Kali Linux 2026.1. Every step below includes the exact command used, the result I observed, a screenshot as evidence, and a short note on why the finding matters from an attacker's point of view.

3. Tools Used

The table below lists each tool used in this report and its purpose.

Tool	Purpose
WHOIS	Domain registration and ownership information gathering.
WhatWeb	Web technology and server fingerprinting (CMS, plugins, IP).
Nslookup	DNS resolution and IP address discovery.
Curl	HTTP header and server information gathering.
Wafw00f	Web Application Firewall detection.
DNSRecon	Enumerate all DNS records (NS, MX, SPF, TXT, SRV).
Google Hacking (GHDB)	Find exposed cameras and downloadable academic PDF documents using search dorks.
Maltego	Visual OSINT and relationship mapping.
theHarvester	Passive Reconnaissance and OSINT collection.
Zenmap	Network discovery and port scanning.
Kali Linux	Operating systems used for reconnaissance and scanning activities.
Kali Linux CMD (Ip route)	Local IP address, interface, route and gateway identification.

4. Activities Performed

a. Footprinting & Reconnaissance (W2-PM1)

I performed reconnaissance against the networkwalks.com domain using six Kali Linux tools: **WHOIS**, **WhatWeb**, **Nslookup**, **Curl**, **Wafw00f**, and **DNSRecon**.

- * **WHOIS:** Queried domain registration details, revealing GoDaddy as registrar, active dates (2019–2027), HostGator nameservers (ns6135.hostgator.com, ns6136.hostgator.com), and WHOIS privacy protection.
- * **WhatWeb:** Identified WordPress 7.0.4, WP Download Manager 3.3.58, Apache web server, jQuery 3.7.1, Bootstrap 7.0.4, and IP address 192.232.216.135.
- * **Nslookup:** Resolved networkwalks.com to its direct host IP 192.232.216.135.
- * **Curl -I:** Retrieved HTTP/2 200 response headers, exposing Apache server, caching headers (x-nginx-cache: WordPress), and the WordPress REST API endpoint /wp-json/.
- * **Wafw00f:** Detected that the website is protected by **ModSecurity (SpiderLabs) WAF**. The captured output also showed several HTTP response signatures, including: • 404 • 405 • 403 • 502 • 500. These responses were observed during WAF fingerprinting and do not by themselves indicate vulnerabilities.
- * **DNSRecon:** Enumerated DNS records including NS, SOA, MX (mail.networkwalks.com), SPF records, cPanel autodiscover SRV records, and exposed BIND version 9.16.23-RH.

b. Search Engine Footprinting & GHDB (W2-PM2)

I used Google search operators (`intitle`, `inurl`, `site`, `filetype`) to locate publicly accessible camera feeds and downloadable academic mathematics resources (IP endpoints partially sanitized):

10x Live Vulnerable / Exposed Security Cameras (Sanitized Endpoints)

No.	Target Link (Sanitized)	Relevant Dork	Access Status
1	http://122.116.41.xxx:8080/	intitle:"webcamXP" inurl:8080	Open webcamXP live stream
2	https://www.lmc.edu/webcam.htm	intitle:"Webcam" inurl:WebCam.htm	Public campus webcam
3	http://198.41.49.xxx:81/main.htm	intitle:"Device(IP CAMERA)" "language" -com net	Direct IP camera stream
4	http://86.122.80.xxx/Pages/login.htm	intitle:"NoVus IP camera" -com	NoVus camera login interface
5	https://www.skylinewebcams.com/en/webcam/...	inurl:webcam site:skylinewebcams.com inurl:roma	Public live broadcast feed
6	https://www.skylinewebcams.com/en/webcam/...	inurl:webcam site:skylinewebcams.com inurl:roma	Public live broadcast feed
7	http://109.233.191.xxx:8080/multi.html	intitle:"webcamXP" inurl:8080	Multi-channel webcamXP stream
8	http://72.199.200.xxx:8080/	intitle:"Index of" "DCIM/camera"	Open directory with camera files
9	http://139.64.168.xxx:8080/	intitle:"Index of" "DCIM/camera"	Open directory camera media storage
10	http://75.149.26.xxx:1024/	intitle:"webcamXP" inurl:8080	webcamXP stream on port 1024

10x Downloadable Mathematics Ebooks / Lecture Notes

No.	Link	Relevant Dork	Institution / Topic
1	https://www.skylineuniversity.ac.ae/pdf/math/	intitle:index.of "parent directory" mathematics pdf	Skyline University (Math Directory)
2	https://www.math.k-state.edu/~gerald/math220d/lec1.pdf	site:.edu filetype:pdf "calculus" "lecture notes"	Kansas State University (Calculus)
3	https://empslocal.ex.ac.uk/people/staff/mrwatkin/zeta/knauf1.pdf	site:.ac.uk ext:pdf "number theory" "introduction"	University of Exeter (Number Theory)
4	https://math.nd.edu/assets/150763/60610_basic_discrete_mathematics.pdf	site:math.*.edu filetype:pdf "discrete mathematics"	Notre Dame (Discrete Mathematics)
5	https://www.maths.usyd.edu.au/u/UG/HM/coordinator/applied2025.pdf	site:.edu.au ext:pdf "applied mathematics"	University of Sydney (Applied Math)
6	https://cas.minesparis.psl.eu/~rouchon/publications/PR1993/INDEXLIN.pdf	intitle:"index of" "linear algebra" pdf	Mines Paris (Linear Algebra)
7	https://people.tamu.edu/~e-straube/Math618/syllabusFall2024.pdf	inurl:syllabus filetype:pdf "complex variables"	Texas A&M (Complex Variables)
8	https://ramanujan.math.trinity.edu/wtrench/texts/TRENCH_REAL_ANALYSIS.PDF	ext:pdf inurl:course "real analysis"	Trinity University (Real Analysis)
9	https://math.njit.edu/sites/math/files/Math_279-001-003-F20.pdf	inurl:downloads filetype:pdf "statistics and probability"	NJIT (Statistics & Probability)
10	https://mrcet.com/downloads/digital_notes/ME/II%20year/MATERIAL%20SCIENCE.pdf	inurl:materials filetype:pdf "geometry"	MRCET Digital Notes

c. Maltego Visual OSINT (W2-PM3)

Using Maltego Graph 4.12.1, I performed visual entity mapping and relationship analysis:

- * Configured Maltego Data Hub transforms.
- * Ran email transforms against networkwalks.com to confirm domain contacts (info@networkwalks.com).

d. theHarvester Footprinting (W2-PM4)

Using theHarvester 4.10.1, I conducted passive subdomain and email discovery. Ran theHarvester -d networkwalks.com -l 1000 -b all. Discovering:

- * 3 ASNs identified (AS13335 / Cloudflare, AS31898, AS46606)
- * 2 public IPs: 172.67.xxx.xxx, 192.232.xxx.xxx (masked — Cloudflare / hosting provider range, full addresses in report)
- * 1 public contact email: info@networkwalks.com
- * 23 subdomain/host records (cpanel, webmail, autodiscover, mail, ftp, etc.)

e. Network Scanning with Zenmap (W2-PM5)

For the internal scanning activity, I performed host discovery and port enumeration on my local LAN (192.168.x.x/24).

- * Ran `ip route` on Kali Linux to identify my local IP address 192.168.x.xxx/24, subnet mask /24 (255.255.xxx.0), and default gateway 192.168.0.1 through the wlan0 interface.
- * Executed `nmap -T4 -F 192.168.0.xxx/24` in Zenmap.
- * Discovered 3 active hosts on the network:
 - 192.168.0.1 (Gateway/Router): Ports **53/tcp (DNS)** and **80/tcp (HTTP)** open (MAC address sanitized).
 - 192.168.x.xxx: **77/tcp filtered**; service was not identified (MAC address sanitized).
 - 192.168.x.xxx (Kali Linux Workstation): Host is up, but all 100 scanned TCP ports were **filtered** with no response.

5. Risk Analysis / Impact

Based on the information collected during the footprinting and network scanning activities, I identified the following potential risks:

#	Risk / Finding	Evidence	Potential Impact	Risk Level
1	Web technology information exposed	WhatWeb found WordPress 7.1 and WP Download Manager 3.3.58	Attackers can search for CVEs in these specific versions	● Medium
2	BIND version exposed on nameservers	DNSRecon found BIND 9.16.23-RH on both nameservers	Version disclosure assists targeting specific server vulnerabilities	● Medium
3	WordPress REST API endpoint exposed	Curl found /wp-json/ in HTTP headers	API can enumerate usernames and post content	● Medium
4	21 subdomains discovered passively	theHarvester found cpanel, webmail, webdisk, autodiscover	Subdomains reveal hosting infrastructure and attack surfaces	● Medium
5	Exposed network cameras found via GHDB	10 live cameras located via Google dorks	Serious privacy and physical security risk	● High
6	Server IP directly identifiable	Nslookup resolved domain to 192.232.216.135	Direct network location of the web server exposed	● Low
7	WAF technology identifiable	Wafw00f found ModSecurity (SpiderLabs)	Reveals perimeter security architecture to attackers	● Low
8	Router HTTP port open on local network	Zenmap found port 80 open on 192.168.0.1	Router admin panel potentially accessible from local network	● Medium
9	Email address harvested passively	theHarvester and Maltego found info@networkwalks.com	Email could be targeted in phishing or social engineering	● Low

Risk Level Key: ● Critical | ● Medium | ● Low

The risks above are observations from the footprinting and scanning exercises, not confirmed vulnerabilities. The practical exercises primarily involved information gathering and host discovery. No exploitation or vulnerability validation was performed as part of these modules.

Therefore, the presence of information such as a software version, IP address, or DNS record does not by itself mean that the system is vulnerable. Further authorized security testing would be required to confirm any actual vulnerability.

6. Recommendations

Based on the observations from these activities, I recommend the following security improvements:

1. **Review publicly exposed technology information:** Regularly check what web technologies, CMS versions, and plugins are publicly visible.
2. **Keep software updated:** Ensure WordPress core, plugins, and web servers are routinely patched.
3. **Review HTTP headers:** Suppress unnecessary server banners and add security headers (HSTS, CSP, X-Frame-Options).
4. **Review DNS records regularly:** Periodically audit DNS and suppress BIND version banner leakage.
5. **Properly configure and monitor the WAF:** Keep ModSecurity enabled and tuned with updated rule sets.
6. **Perform regular internal network discovery:** Periodically scan internal subnets to identify unauthorized or rogue devices.
7. **Secure internal SMB and NetBIOS:** Restrict inbound access to ports 135, 139, and 445 on local workstations.
8. **Disable UPnP on gateway devices:** Turn off UPnP on routers to prevent automated port forwarding.
9. **Maintain network documentation:** Keep network topology diagrams and IP address assignments updated.
10. **Perform security testing with authorization:** Always ensure scanning and OSINT testing are conducted within an authorized scope.

7. Conclusion

During Week 2 of my Cybersecurity internship at NetworkWalks Academy, I completed practical activities covering six areas of footprinting, OSINT, and network scanning.

In the footprinting activity (W2-PM1), I used six Kali Linux tools to collect information about networkwalks.com. I learned how WHOIS reveals domain ownership, WhatWeb identifies web technologies, Nslookup resolves domain names, Curl inspects HTTP headers, Wafw00f identifies WAFs, and DNSRecon enumerates DNS infrastructure.

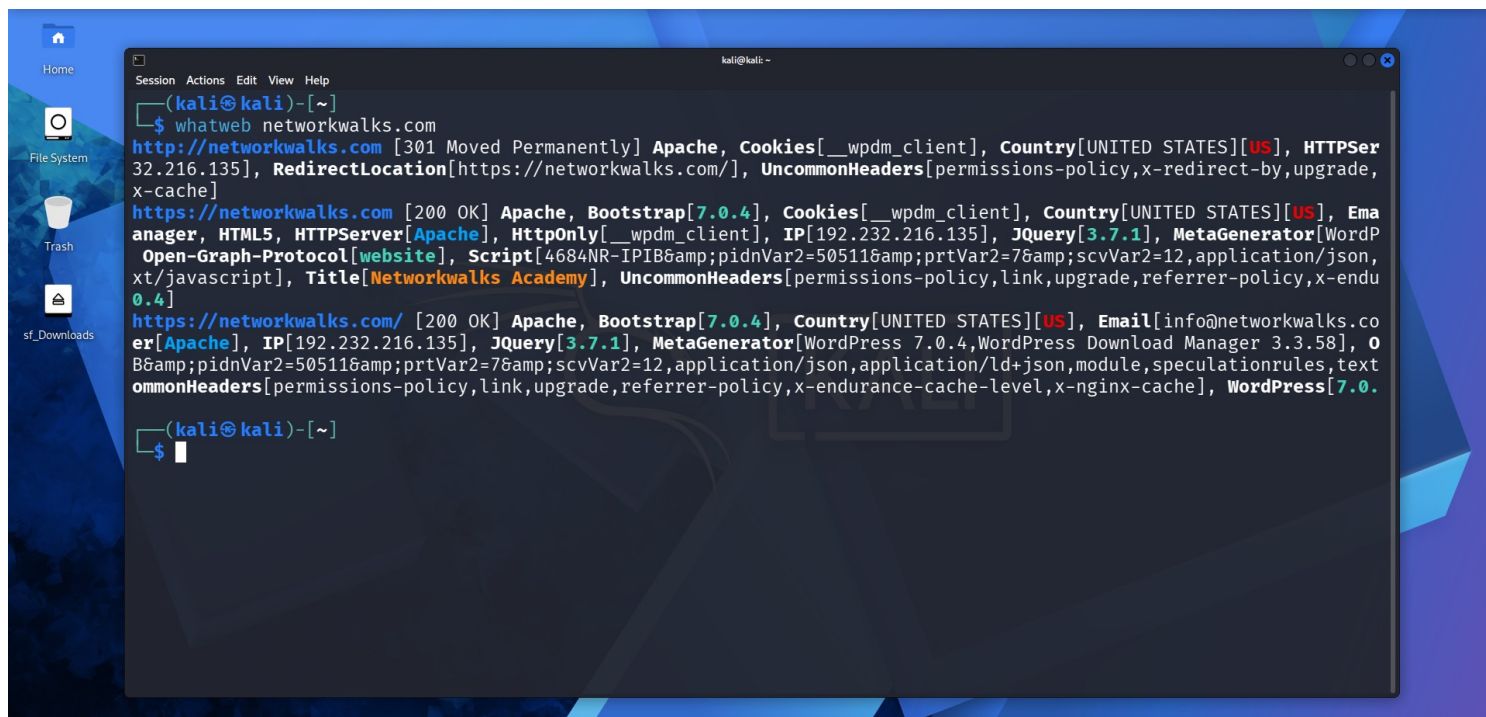
In the Google Dorking activity (W2-PM2), I discovered how powerful search operators can uncover publicly indexed sensitive resources including live cameras and open file directories — all without touching the target directly.

In the Maltego activity (W2-PM3), I saw how visual graph analysis connects domain entities to email addresses and other infrastructure nodes, making relationships easy to see at a glance.

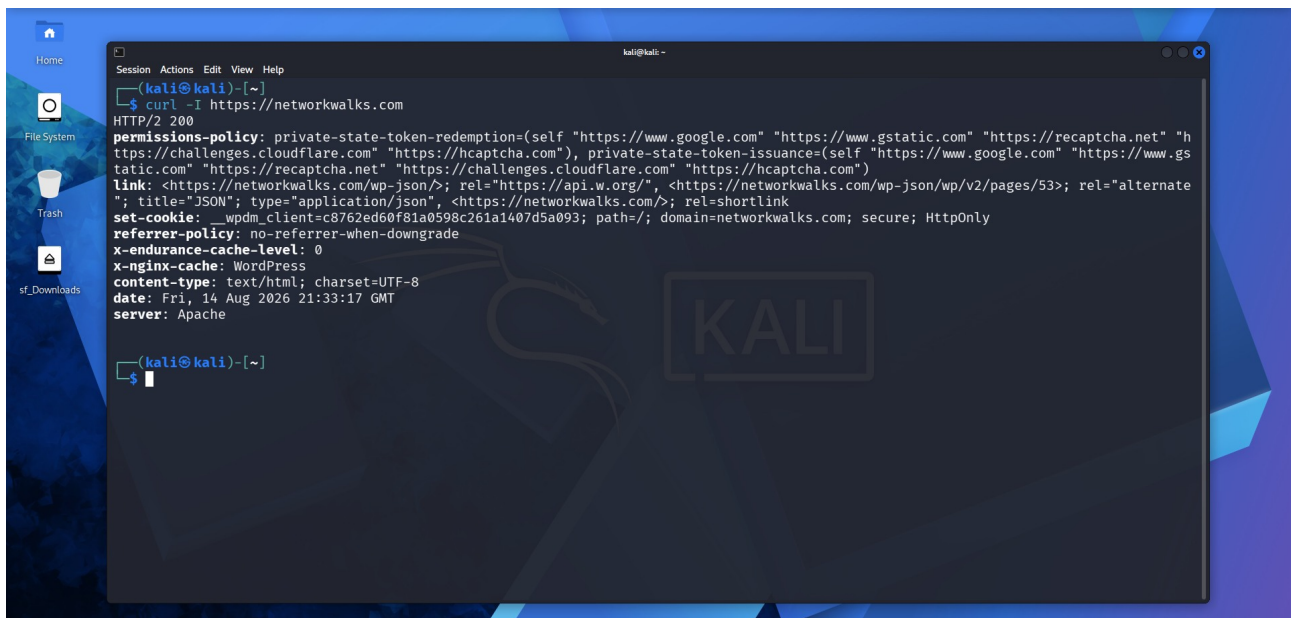
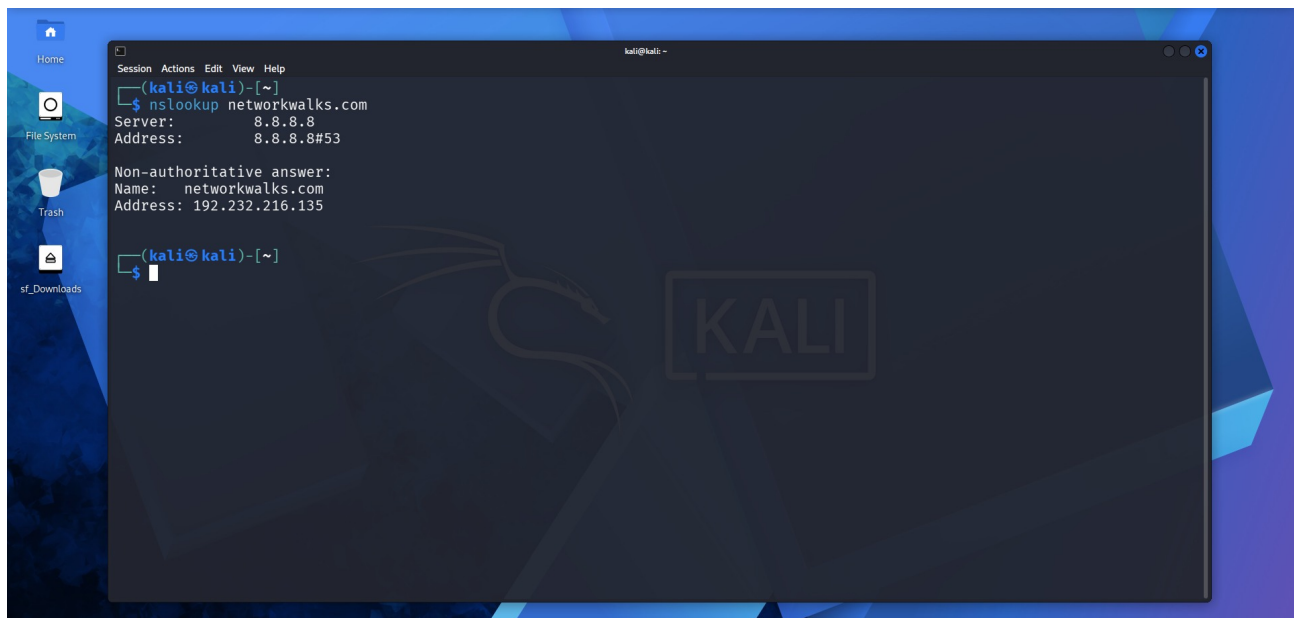
In the theHarvester activity (W2-PM4), I collected 21 subdomains, 2 IPs, and 1 email address from networkwalks.com completely passively, showing that a significant amount of intelligence can be gathered before any active scanning begins.

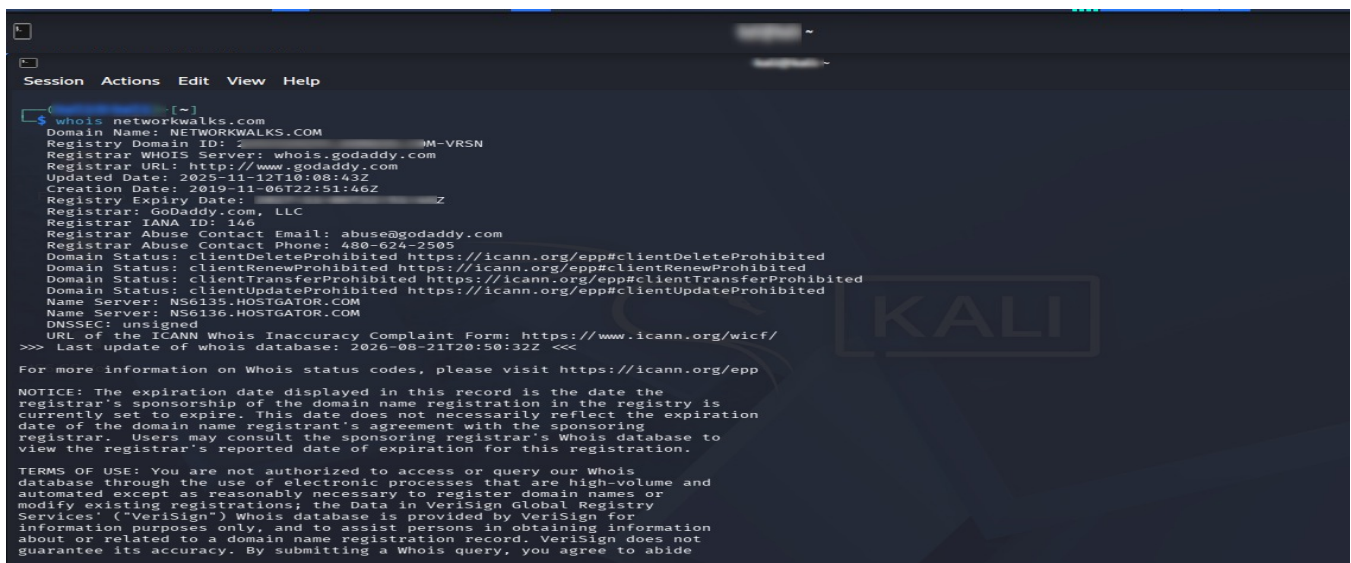
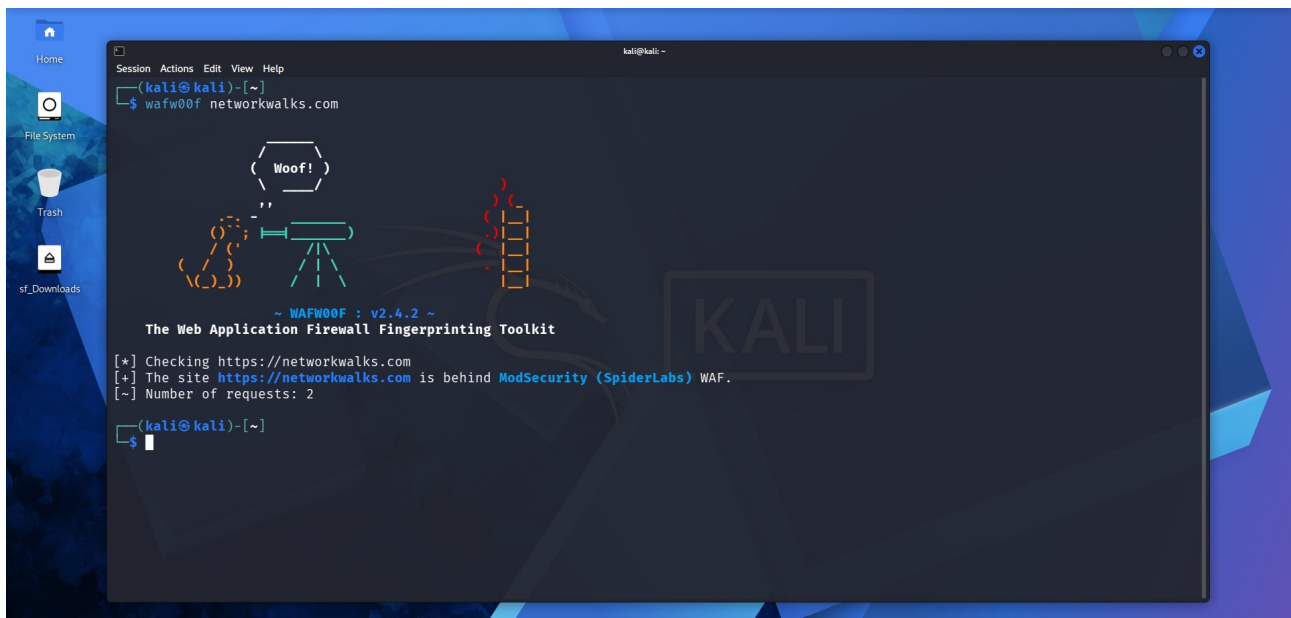
In the Zenmap activity (W2-PM5), I mapped my local network and discovered 3 live hosts with their open ports and MAC addresses, learning how network topology can be quickly understood through automated scanning.

8. Evidences Collected



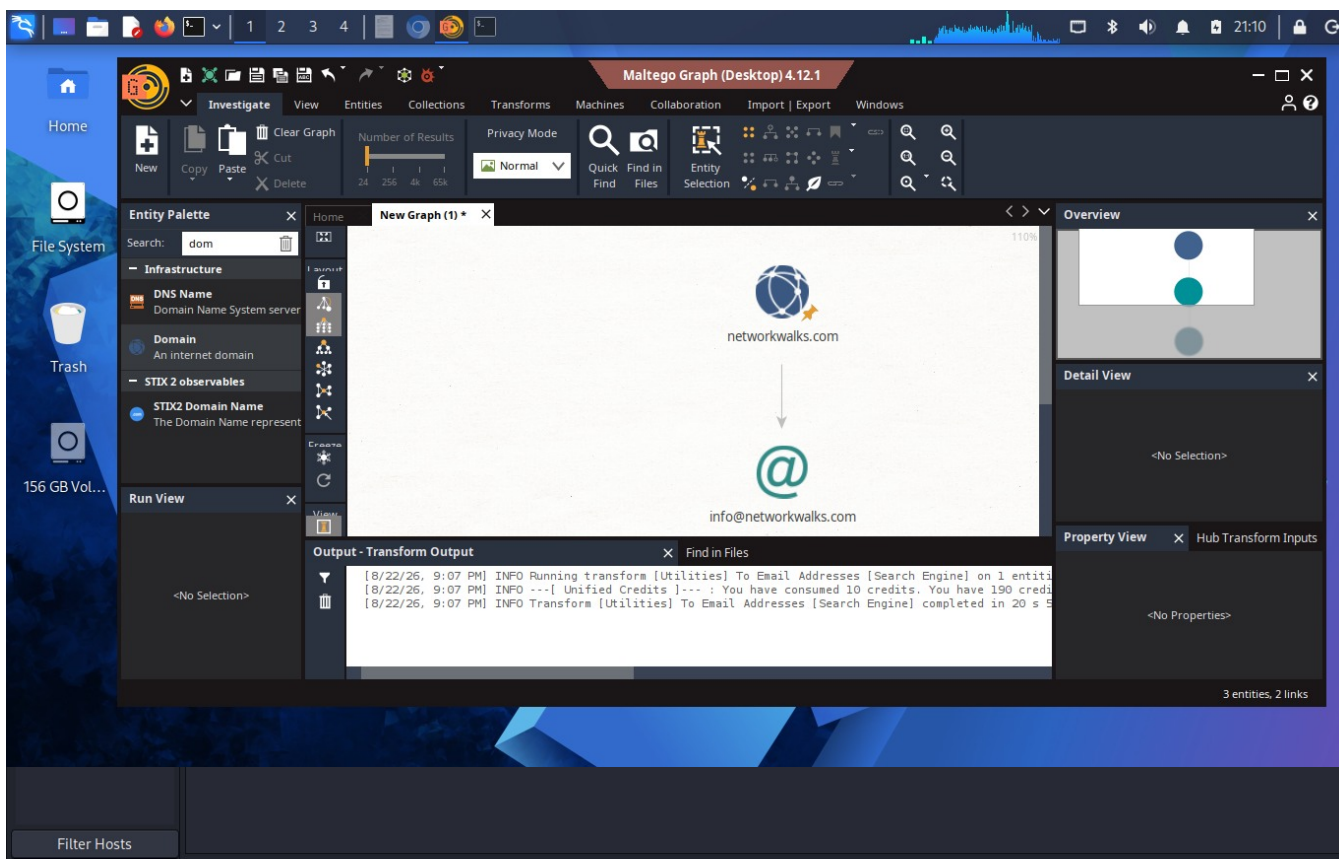
```
kali@kali: ~  
$ whatweb networkwalks.com  
http://networkwalks.com [301 Moved Permanently] Apache, Cookies[__w...  
https://networkwalks.com [200 OK] Apache, Bootstrap[7.0.4], Cookies[__w...  
https://networkwalks.com/ [200 OK] Apache, Bootstrap[7.0.4], Country[UNITED STATES][US], Email[info@networkwalks.co...  
er[Apache], IP[192.232.216.135], JQuery[3.7.1], MetaGenerator[WordPress 7.0.4,WordPress Download Manager 3.3.58], O...  
B&pidnVar2=50511&prtVar2=7&scvVar2=12,application/json,application/ld+json,module,speculationrules,text...  
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```

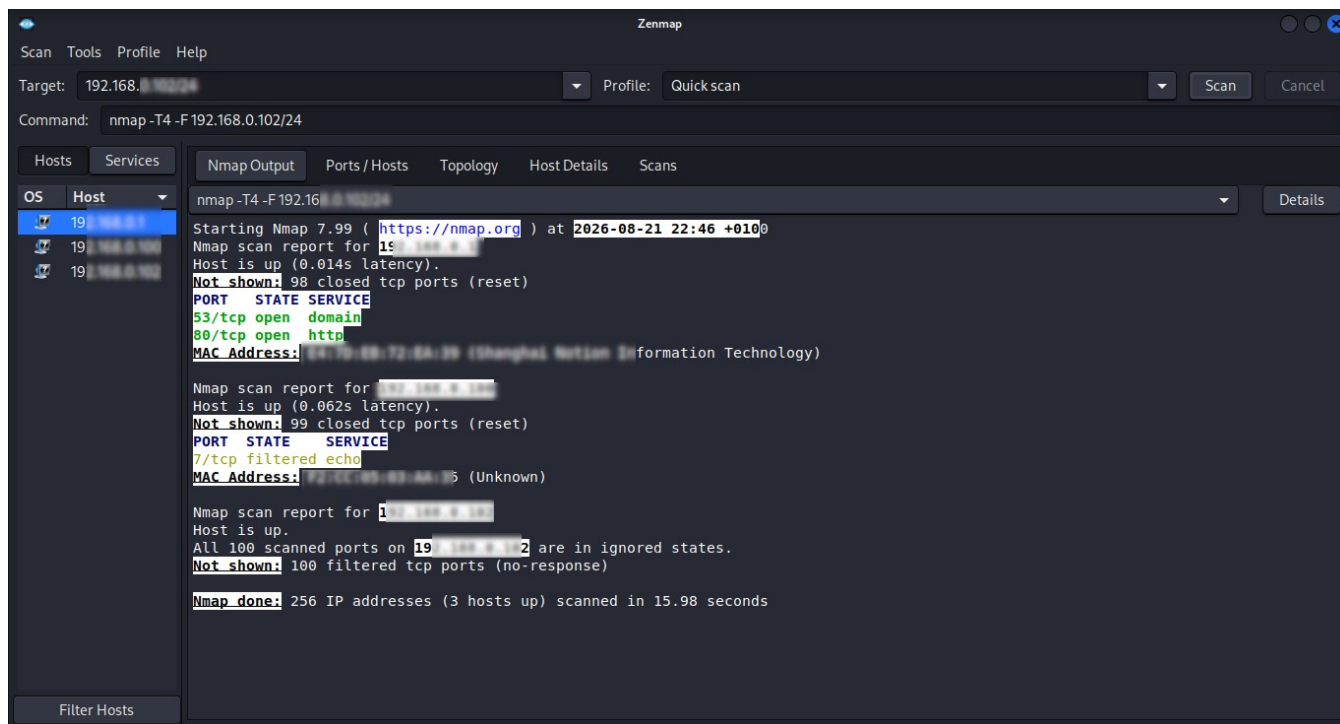





```
Session Actions Edit View Help
(ka [redacted])-[~]
$ dnsrecon -d networkwalks.com
2026-08-21T22:23:44.114630+0100 INFO Starting enumeration for domain: networkwalks.com
2026-08-21T22:23:44.115064+0100 INFO std: Performing General Enumeration against: networkwalks.com ...
2026-08-21T22:23:45.335146+0100 ERROR No answer for DNSSEC query for networkwalks.com
2026-08-21T22:23:46.281597+0100 INFO SOA ns6135.hostgator.com 50.87.144.87
2026-08-21T22:23:47.485696+0100 INFO NS ns6136.hostgator.com 192.232.216.131
2026-08-21T22:23:48.613193+0100 INFO Bind Version for 192.232.216.131 "9.16.23-RH"
2026-08-21T22:23:48.613193+0100 INFO NS ns6135.hostgator.com 50.87.144.87
2026-08-21T22:23:49.567240+0100 INFO Bind Version for 50.87.144.87 "9.16.23-RH"
2026-08-21T22:23:50.148823+0100 INFO MX mail.networkwalks.com 1
2026-08-21T22:23:50.892156+0100 INFO A networkwalks.com 192.232.216.131
2026-08-21T22:23:52.092703+0100 INFO TXT networkwalks.com v=spf1 +a +mx +ip4:50.87.144.87 +include:websitewelcome.com ~all
2026-08-21T22:23:52.092959+0100 INFO TXT networkwalks.com google-site-verification=rr04eRmqHoWY3XemnizDNVK4q75X-Ij-mjgEeg-UsYI
2026-08-21T22:23:53.219342+0100 INFO Enumerating SRV Records
2026-08-21T22:24:02.603685+0100 ERROR No SRV Records Found for networkwalks.com
2026-08-21T22:24:02.604981+0100 INFO Completed enumeration for domain: networkwalks.com

(ka [redacted])-[~]
```





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